

Hilbert-Valued Characteristic Domination for All Finite p

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2026-06-16

Status

This packet gives a candidate partial result, likely valid, for Remark 12.10 of Yaroslavtsev's paper *Local characteristics and tangency of vector-valued martingales* [1]. It solves the Hilbert-valued subcase of the general-local-martingale characteristic-domination problem for every $1 \leq p < \infty$. It is partial relative to the broad source question because it does not cover non-Hilbert UMD spaces.

The new part of this packet is the range $1 \leq p < 2$. The already packaged Hilbert $p = 2$ and $p \geq 2$ packets handled the upper range by the Hilbert square identity and Rosenthal/BDG estimates. Below 2, the missing substitute is a Laplace-transform comparison for random square functions.

Source Problem

The source is arXiv:1907.11588, PDF page 105, Remark 12.10. The source crop below states that the general-local-martingale case is open and reduces the missing accessible-jump issue to a discrete independent-increment comparison.

...and the fact that M and N are quasi-left continuous. $\square$

Remark 12.10. It is not known whether Theorem 12.8 holds for general local martingales. By Theorem 3.39 and by Proposition B.1 the main issue here is in proving a similar statement for discrete martingales with independent increments. Let us state this problem here as open. Let X be a Banach space, $1 \leq p < \infty$. Let $(\xi_n)_{n \geq 1}$ and $(\eta_n)_{n \geq 1}$ be X -valued mean-zero independent random variables such that for any Borel $B \in X \setminus \{0\}$

$$\sum_n \mathbb{P}(\xi_n \in B) \leq \sum_n \mathbb{P}(\eta_n \in B).$$

Does there exists a constant C (perhaps depending on p and X) such that

$$\mathbb{E} \left\| \sum_n \xi_n \right\|^p \leq C \mathbb{E} \left\| \sum_n \eta_n \right\|^p?$$

The next page adds that, by symmetrization, one may assume the variables are symmetric, and that even this symmetric discrete case was not known to the author.

By a standard symmetrization trick [66, Lemma 6.3] one can assume that ξ_n 's and η_n 's are symmetric. But even the symmetric case is not known for the author.

In discrete form, the source asks whether independent mean-zero X -valued variables $(\xi_n), (\eta_n)$ satisfying

$$\sum_n \mathbb{P}(\xi_n \in B) \leq \sum_n \mathbb{P}(\eta_n \in B) \quad (B \subset X \setminus \{0\} \text{ Borel})$$

must satisfy

$$\mathbb{E} \left\| \sum_n \xi_n \right\|^p \leq C_{p,X} \mathbb{E} \left\| \sum_n \eta_n \right\|^p.$$

Main Result

Theorem 1. *Let H be a separable Hilbert space and $1 \leq p < \infty$. Let M and N be H -valued local martingales such that N is characteristically dominated by M in the sense of [1]. Then*

$$\mathbb{E} \sup_{t \geq 0} \|N_t\|^p \leq C_p \mathbb{E} \sup_{t \geq 0} \|M_t\|^p,$$

whenever the right hand side is finite.

Idea of the Proof

The hard range is $1 \leq p < 2$. A naive Rosenthal estimate is poorly matched there: the term $\sum_n \mathbb{E} \|Z_n\|^p$ is too large and is not comparable to the p -moment of the sum. The right object is instead the random square function

$$S_Z = \left(\sum_n \|Z_n\|^2 \right)^{1/2}.$$

For independent mean-zero Hilbert-valued variables,

$$\mathbb{E} \left\| \sum_n Z_n \right\|^p \simeq_p \mathbb{E} S_Z^p, \quad 1 \leq p < \infty.$$

Thus the discrete problem below 2 becomes a scalar question: if the aggregate law of the nonnegative variables $Y_n = \|\xi_n\|^2$ is dominated by that of $V_n = \|\eta_n\|^2$, does

$$\mathbb{E} \left(\sum_n Y_n \right)^\alpha \lesssim_\alpha \mathbb{E} \left(\sum_n V_n \right)^\alpha, \quad \alpha = p/2 < 1,$$

hold? Yes. For $S = \sum_n Y_n$,

$$\mathbb{E} S^\alpha = c_\alpha \int_0^\infty (1 - \mathbb{E} e^{-tS}) t^{-1-\alpha} dt,$$

and independence gives

$$1 - \mathbb{E}e^{-tS} = 1 - \prod_n (1 - u_n(t)), \quad u_n(t) = \mathbb{E}(1 - e^{-tY_n}).$$

Writing $U(t) = \sum_n u_n(t)$, one has

$$(1 - e^{-1}) \min\{1, U(t)\} \leq 1 - \prod_n (1 - u_n(t)) \leq \min\{1, U(t)\}.$$

The quantity $U(t)$ depends only on the aggregate law. Hence aggregate domination controls every fractional square-function moment with $\alpha < 1$. This is the below-2 good-lambda substitute.

Discrete Hilbert Comparison

Lemma 1 (Independent Hilbert sums and square functions). *Let H be a Hilbert space, $1 \leq p < \infty$, and let $Z_1, \dots, Z_N$ be independent mean-zero H -valued random variables. Then*

$$\mathbb{E} \left\| \sum_{n=1}^N Z_n \right\|^p \simeq_p \mathbb{E} \left(\sum_{n=1}^N \|Z_n\|^2 \right)^{p/2}.$$

Moreover,

$$\mathbb{E} \max_{1 \leq m \leq N} \left\| \sum_{n=1}^m Z_n \right\|^p \simeq_p \mathbb{E} \left(\sum_{n=1}^N \|Z_n\|^2 \right)^{p/2}.$$

Proof. The maximal estimate is the classical Hilbert-valued discrete Burkholder-Davis-Gundy inequality, recalled for instance in [2]. Since H is Hilbert, the Gaussian/radonifying term in the discrete BDG theorem is exactly $(\sum_n \|Z_n\|^2)^{1/2}$.

For the terminal estimate, let $Z'_1, \dots, Z'_N$ be an independent copy and write $S = \sum_n Z_n$, $S' = \sum_n Z'_n$, and $W_n = Z_n - Z'_n$. By Jensen,

$$\mathbb{E} \|S\|^p = \mathbb{E} \|\mathbb{E}(S - S' \mid Z_1, \dots, Z_N)\|^p \leq \mathbb{E} \|S - S'\|^p.$$

The variables W_n are independent and symmetric, so their joint law is invariant under independent sign changes. Hence, for an auxiliary Rademacher sequence (ε_n) , the Hilbert-space Khintchine inequality gives

$$\mathbb{E} \|S - S'\|^p = \mathbb{E} \mathbb{E}_\varepsilon \left\| \sum_n \varepsilon_n W_n \right\|^p \simeq_p \mathbb{E} \left(\sum_n \|W_n\|^2 \right)^{p/2} \lesssim_p \mathbb{E} \left(\sum_n \|Z_n\|^2 \right)^{p/2}.$$

For the final inequality we used the triangle inequality in $L^p(\Omega; \ell_2(H))$.

Conversely,

$$\mathbb{E} \|S - S'\|^p \leq 2^p \mathbb{E} \|S\|^p,$$

again by the triangle inequality and the equality in law of S and S' . The lower Khintchine inequality and Jensen applied in the Hilbert space $\ell_2^N(H)$ give

$$\mathbb{E} \|S\|^p \gtrsim_p \mathbb{E} \left(\sum_n \|W_n\|^2 \right)^{p/2} \geq \mathbb{E} \left(\sum_n \|\mathbb{E}(W_n \mid Z_1, \dots, Z_N)\|^2 \right)^{p/2}.$$

Since $\mathbb{E}(W_n \mid Z_1, \dots, Z_N) = Z_n$, this is the desired lower bound. $\square$

Lemma 2 (Laplace comparison for fractional square functions). *Let $0 < \alpha < 1$. For $i = 1, \dots, N$, let Y_i and V_i be independent nonnegative random variables. Let μ_i and ν_i be the restrictions of their laws to $(0, \infty)$. If*

$$\sum_{i=1}^N \mu_i \leq \sum_{i=1}^N \nu_i \quad \text{as measures on } (0, \infty),$$

then

$$\mathbb{E} \left(\sum_{i=1}^N Y_i \right)^\alpha \leq \frac{1}{1 - e^{-1}} \mathbb{E} \left(\sum_{i=1}^N V_i \right)^\alpha.$$

Proof. For $S_Y = \sum_i Y_i$, set

$$u_i(t) = \mathbb{E}(1 - e^{-tY_i}) = \int_{(0, \infty)} (1 - e^{-ty}) d\mu_i(y), \quad U_Y(t) = \sum_i u_i(t).$$

Then $0 \leq u_i(t) \leq 1$, and independence gives

$$\mathbb{E} e^{-tS_Y} = \prod_i (1 - u_i(t)).$$

Therefore

$$1 - \mathbb{E} e^{-tS_Y} = 1 - \prod_i (1 - u_i(t)) \leq \min\{1, U_Y(t)\},$$

while

$$1 - \mathbb{E} e^{-tS_Y} \geq 1 - e^{-U_Y(t)} \geq (1 - e^{-1}) \min\{1, U_Y(t)\}.$$

The same statements hold for $S_V = \sum_i V_i$. The measure domination implies

$$U_Y(t) = \int_{(0, \infty)} (1 - e^{-ty}) d \sum_i \mu_i(y) \leq \int_{(0, \infty)} (1 - e^{-ty}) d \sum_i \nu_i(y) = U_V(t).$$

Hence, for every $t > 0$,

$$1 - \mathbb{E} e^{-tS_Y} \leq \frac{1}{1 - e^{-1}} (1 - \mathbb{E} e^{-tS_V}).$$

Finally use the standard fractional-moment identity

$$\mathbb{E} S^\alpha = \frac{\alpha}{\Gamma(1 - \alpha)} \int_0^\infty (1 - \mathbb{E} e^{-tS}) t^{-1-\alpha} dt \quad (0 < \alpha < 1).$$

□

Proposition 1 (Discrete aggregate domination in Hilbert spaces below 2). *Let H be a Hilbert space and $1 \leq p < 2$. Let $(\xi_n)_{n=1}^N$ and $(\eta_n)_{n=1}^N$ be independent mean-zero H -valued random variables. If*

$$\sum_{n=1}^N \mathbb{P}(\xi_n \in B) \leq \sum_{n=1}^N \mathbb{P}(\eta_n \in B) \quad (B \subset H \setminus \{0\} \text{ Borel}),$$

then

$$\mathbb{E} \left\| \sum_{n=1}^N \xi_n \right\|^p \leq C_p \mathbb{E} \left\| \sum_{n=1}^N \eta_n \right\|^p.$$

The same comparison holds with terminal sums replaced by the corresponding maxima over partial sums in any fixed order.

Proof. Put $Y_n = \|\xi_n\|^2$, $V_n = \|\eta_n\|^2$, and $\alpha = p/2 \in [1/2, 1)$. Applying the map $x \mapsto \|x\|^2$ to the aggregate jump measures gives

$$\sum_n \mathcal{L}(Y_n)|_{(0,\infty)} \leq \sum_n \mathcal{L}(V_n)|_{(0,\infty)}.$$

Lemma 2 yields

$$\mathbb{E} \left(\sum_n \|\xi_n\|^2 \right)^{p/2} \leq C \mathbb{E} \left(\sum_n \|\eta_n\|^2 \right)^{p/2}.$$

The terminal and maximal assertions now follow from Lemma 1. $\square$

Proposition 2 (Discrete aggregate domination in Hilbert spaces for all p). *Let H be a Hilbert space and $1 \leq p < \infty$. Let $(\xi_n)_{n=1}^N$ and $(\eta_n)_{n=1}^N$ be independent mean-zero H -valued random variables satisfying the aggregate domination hypothesis of Proposition 1. Then*

$$\mathbb{E} \left\| \sum_{n=1}^N \xi_n \right\|^p \leq C_p \mathbb{E} \left\| \sum_{n=1}^N \eta_n \right\|^p,$$

and the same comparison holds for maxima over partial sums in any fixed order.

Proof. The case $1 \leq p < 2$ is Proposition 1. Let $p \geq 2$ and set $r = p/2 \geq 1$. Lemma 1 reduces matters to square functions. For $Y_n = \|\xi_n\|^2$, the scalar Rosenthal inequality for independent nonnegative variables gives

$$\mathbb{E} \left(\sum_n Y_n \right)^r \lesssim_p \left(\sum_n \mathbb{E} Y_n \right)^r + \sum_n \mathbb{E} Y_n^r.$$

Aggregate domination controls the two terms:

$$\sum_n \mathbb{E} \|\xi_n\|^2 \leq \sum_n \mathbb{E} \|\eta_n\|^2, \quad \sum_n \mathbb{E} \|\xi_n\|^p \leq \sum_n \mathbb{E} \|\eta_n\|^p.$$

For $V_n = \|\eta_n\|^2$,

$$\left(\sum_n \mathbb{E} V_n \right)^r + \sum_n \mathbb{E} V_n^r \leq 2 \mathbb{E} \left(\sum_n V_n \right)^r,$$

by Jensen and the pointwise inequality $(\sum_n V_n)^r \geq \sum_n V_n^r$. Thus the ξ -square-function p -moment is controlled by the η -square-function p -moment, and Lemma 1 finishes the proof. The endpoint $p = 2$ is also immediate from the Hilbert square identity. $\square$

Independent-Increment Characteristic Comparison

Proposition 3 (Hilbert independent-increment comparison). *Let H be a separable Hilbert space and $1 \leq p < \infty$. Let L and K be H -valued martingales with independent increments and deterministic local characteristics (A_L, ν_L) and (A_K, ν_K) . Assume*

$$\mathrm{tr} A_L(\infty) \leq \mathrm{tr} A_K(\infty), \quad \nu_L(\mathbb{R}_+ \times B) \leq \nu_K(\mathbb{R}_+ \times B)$$

for every Borel $B \subset H \setminus \{0\}$. Then

$$\mathbb{E} \sup_{t \geq 0} \|L_t\|^p \leq C_p \mathbb{E} \sup_{t \geq 0} \|K_t\|^p.$$

Proof. By the Hilbert-valued BDG inequality,

$$\mathbb{E} \sup_{t \geq 0} \|L_t\|^p \simeq_p \mathbb{E} Q_L^{p/2}, \quad Q_L := \operatorname{tr} A_L(\infty) + \sum_{s \geq 0} \|\Delta L_s\|^2,$$

and similarly for K . Thus it is enough to compare the scalar square characteristics Q_L and Q_K .

First suppose $1 \leq p < 2$ and put $\alpha = p/2$. For a martingale with deterministic characteristics (A, ν) , write

$$a = \operatorname{tr} A(\infty), \quad U_{A, \nu}(t) = (1 - e^{-ta}) + \int_{\mathbb{R}_+ \times H} (1 - e^{-t\|x\|^2}) d\nu(s, x).$$

The source paper's Grigelionis representation for independent-increment martingales decomposes the jump measure into a Poisson part over the nonatomic-in-time component of ν and independent single jumps over the time atoms of ν . Consequently the Laplace transform of $Q = a + \sum_s \|\Delta L_s\|^2$ has the form

$$\mathbb{E} e^{-tQ} = e^{-ta} e^{-U_0(t)} \prod_j (1 - u_j(t)),$$

where $U_0(t)$ is the contribution of the nonatomic part and $u_j(t)$ are the atom-time contributions. As in Lemma 2,

$$(1 - e^{-1}) \min\{1, U_{A, \nu}(t)\} \leq 1 - \mathbb{E} e^{-tQ} \leq \min\{1, U_{A, \nu}(t)\}.$$

The assumptions imply $U_{A_L, \nu_L}(t) \leq U_{A_K, \nu_K}(t)$ for all $t > 0$. The fractional-moment identity used in Lemma 2 yields

$$\mathbb{E} Q_L^\alpha \lesssim_\alpha \mathbb{E} Q_K^\alpha.$$

Now let $p \geq 2$ and set $\alpha = p/2 \geq 1$. For the same scalar nonnegative independent-increment square sum, the scalar Rosenthal inequality, applied first to finite truncations of the Poisson and atom-time decomposition and then by monotone convergence, gives

$$\mathbb{E} Q^\alpha \lesssim_\alpha \left(a + \int \|x\|^2 d\nu \right)^\alpha + \int \|x\|^{2\alpha} d\nu.$$

The reverse lower bounds needed for K are elementary:

$$\mathbb{E} Q_K^\alpha \geq (\mathbb{E} Q_K)^\alpha = \left(a_K + \int \|x\|^2 d\nu_K \right)^\alpha$$

by Jensen, and

$$\mathbb{E} Q_K^\alpha \geq \mathbb{E} \sum_s \|\Delta K_s\|^{2\alpha} = \int \|x\|^{2\alpha} d\nu_K,$$

because $r \mapsto r^\alpha$ is superadditive on $\mathbb{R}_+$. The assumptions control the two upper-bound terms for L by the corresponding terms for K . Hence $\mathbb{E} Q_L^\alpha \lesssim_\alpha \mathbb{E} Q_K^\alpha$, and BDG completes the proof. $\square$

General Local Martingales

Proof of Theorem 1. Since Hilbert spaces are UMD, Theorem `thm:mainforDTMgencasenanananana` of [1] gives decoupled tangent local martingales $\widetilde{M}$ and $\widetilde{N}$ for M and N . The tangent estimate `thm:tangentgencaseUMDuhoditrotasoldat` gives

$$\mathbb{E} \sup_t \|M_t\|^p \simeq_p \mathbb{E} \sup_t \|\widetilde{M}_t\|^p, \quad \mathbb{E} \sup_t \|N_t\|^p \simeq_p \mathbb{E} \sup_t \|\widetilde{N}_t\|^p.$$

The same source theorem constructs the decoupled tangent process so that, conditional on the original local characteristics, $\widetilde{M}$ and $\widetilde{N}$ have independent increments and deterministic local characteristics $(\llbracket M^c \rrbracket, \nu^M)$ and $(\llbracket N^c \rrbracket, \nu^N)$, respectively.

Fix a base point ω outside the exceptional null sets. Characteristic domination says

$$\llbracket N^c \rrbracket_\infty(\omega) \leq \llbracket M^c \rrbracket_\infty(\omega), \quad \nu^N(\omega; \mathbb{R}_+ \times B) \leq \nu^M(\omega; \mathbb{R}_+ \times B)$$

for every Borel $B \subset H \setminus \{0\}$. The bilinear-form order implies $\text{tr} \llbracket N^c \rrbracket_\infty(\omega) \leq \text{tr} \llbracket M^c \rrbracket_\infty(\omega)$. Proposition 3, applied on this conditional probability space, therefore gives

$$\mathbb{E} \left(\sup_t \left\| \widetilde{N}_t \right\|^p \mid \llbracket N^c \rrbracket, \nu^N, \llbracket M^c \rrbracket, \nu^M \right) \leq C_p \mathbb{E} \left(\sup_t \left\| \widetilde{M}_t \right\|^p \mid \llbracket N^c \rrbracket, \nu^N, \llbracket M^c \rrbracket, \nu^M \right).$$

Integrating over ω and using the tangent estimates above yields the claimed inequality. $\square$

Verification Notes

The only genuinely new step is Lemma 2. It is insensitive to the partition of the aggregate measure into independent summands: all such partitions have Laplace deficits comparable to $\min\{1, U(t)\}$, where $U(t)$ is determined by the aggregate law alone. This is why rare huge jumps do not break the $p < 2$ Hilbert comparison.

The proof does not use symmetry of the original variables. Symmetrization is used only inside Lemma 1, via an independent copy. The mean-zero assumption is exactly what makes $\mathbb{E}(Z_n - Z'_n \mid Z_1, \dots, Z_N) = Z_n$.

The line-by-line reduction audit found one important point: characteristic domination in [1] controls the total spatial jump compensator $\nu(\mathbb{R}_+ \times \cdot)$, not the accessible and quasi-left-continuous compensator parts separately. Thus the general proof above deliberately does not claim that the accessible part of N is dominated by the accessible part of M . Instead it uses the source's full decoupled tangent theorem and compares the resulting independent-increment martingales by their full characteristics in Proposition 3. This is exactly the source reduction, but without imposing an unjustified componentwise domination.

A small verification script, `code/check_laplace_partition.py`, tests random finite partitions of the same aggregate nonnegative law and checks the Laplace-deficit bounds used in Lemma 2.

Novelty and Literature Check

The cheap run indexes were searched for the arXiv id and for the terms **characteristically dominated**, **aggregate domination**, **Hilbert**, **p<2**, **all p**, **Rosenthal**, and **Laplace**. The run already contained the broad c_0 counterexample, the Hilbert $p = 2$ packet, and the Hilbert $p \geq 2$ packet, but no Hilbert $1 \leq p < 2$ or all- p packet. Literature/source searches identified the source reduction in [1], Hilbert-valued BDG and square-function estimates in [2], and Rosenthal/comparison tools such as [3, 4], but no exact recorded solution of the Hilbert $1 \leq p < 2$ subcase of Remark 12.10.

Limitations

This result solves the Hilbert-valued subcase, not the full UMD-space problem. The previous c_0 counterexample shows that the unrestricted Banach-space discrete statement in Remark 12.10 is

false. The remaining interesting direction is therefore geometric: identify which non-Hilbert UMD spaces retain enough square-function or tail comparability to replace Lemma 1.

Human Review Recommendation

First verify Lemma 2; it is short and is the decisive new below-2 mechanism. Then verify Lemma 1, especially the terminal lower bound using the independent copy. Finally verify Proposition 3, which is the hardened replacement for the informal accessible-jump reduction: it compares the full independent- increment characteristics after decoupling, so no separate domination of canonical jump parts is needed.

References

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